Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “diffraction” and “wavelength” to explain your model or result. Sentence starters are optional.

diffraction: wave spreading around an edge or through an opening

wavelength: distance between corresponding points on repeated waves

boundary: the place where a medium or constraint changes

01 Warm-up: bring back an earlier idea

A wave has frequency 4 Hz and wavelength 6 m.
Find its speed.

Word help: frequency = number of cycles each second

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

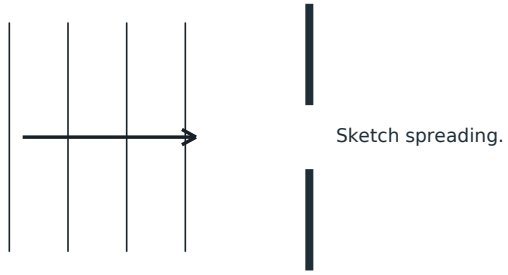
02 Worked example: explain the model

Reference: λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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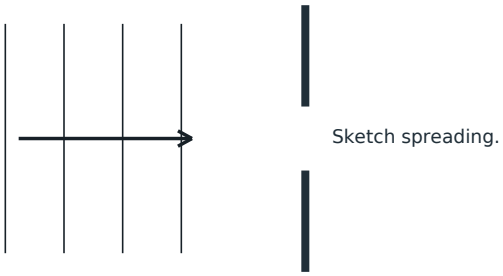
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch wavefronts after a narrow opening about one wavelength wide.
Compare with a much wider opening.

Sentence starter: Both … . They differ in … because … .

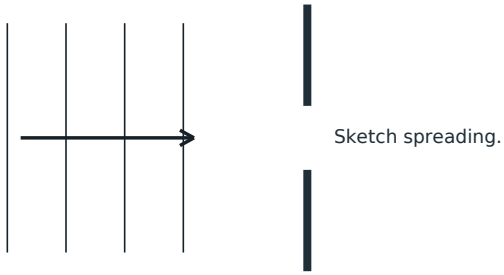


Marzano check - circle: 1 2 3

04 Apply the physics

A ripple-tank simulation sends 1 m wavelength waves toward two openings: A is 1 m wide and B is 10 m wide. **Sketch** the emerging wavefronts. Which opening produces more noticeable spreading relative to its width?

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

05 Explain your reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

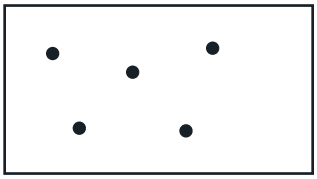
06 Review: connect an earlier idea

At the same kelvin temperature, sample B contains 4 times as many particles of the same ideal gas as A.
Compare average energy per particle and total translational energy.
Use particle arrows and labels to show the comparison.

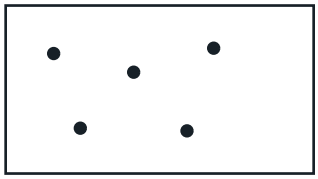
Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



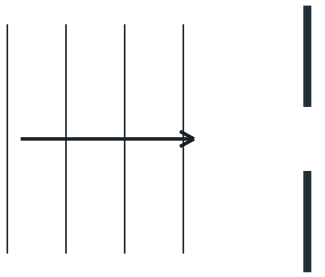
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

For the same opening, would a longer wavelength generally spread more or less?

Sentence starter: I notice … . This supports … because … .



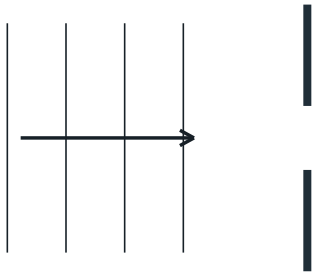
Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep opening width fixed and halve wavelength 0.
Predict the change in relative spreading and draw before/after patterns.

Sentence starter: I predict … . My model supports this because … .



Sketch spreading.

Marzano check - circle: 1 2 3